

# The dynamics of quantum scattering and confinement in 1D & 3D geometries for time-independent potentials

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MIDTERM PROJECT

Applied Mathematics for Engineering



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# Abstract and Motivation

Scattering theory provides a setting in which several standard mathematical tools like ordinary differential equations, special functions, Fourier methods, and complex analysis appear together in a natural and unified way. Rather than separating physical situations into “bound” and “unbound” problems, the same underlying framework can be used to study how wavefunctions behave both near regions of interaction and far away from them. The central idea is to treat the motion of a particle as a propagation problem: given an incident wave, determine how it evolves under a potential, and understand how its structure changes between the interaction region and the asymptotic regime. In this sense, confinement and scattering are not distinct categories but different manifestations of the same dynamics viewed under different boundary conditions.

At the level of the Schrödinger equation, the problem reduces to solving a second-order linear differential equation in regions of constant or slowly varying potential, and enforcing continuity conditions at interfaces. In one dimension this leads directly to reflection and transmission amplitudes, while in three dimensions separation of variables produces a radial equation with an effective centrifugal term. The resulting solutions are expressed in terms of Bessel and spherical Bessel functions, and form the basis of the partial wave decomposition.

Fourier methods enter through the representation of incident states as superpositions of plane waves, making explicit the relation between momentum space and position space descriptions. Extending the analysis to complex wave numbers reveals further structure: poles of the scattering amplitudes correspond to bound states and resonances, providing a direct link between analytic properties and physical spectra.

The aim of this project is to use scattering as a framework in which these techniques are not introduced separately, but seen operating together in a single, coherent construction.

# OUTLINE

## 1. Introduction

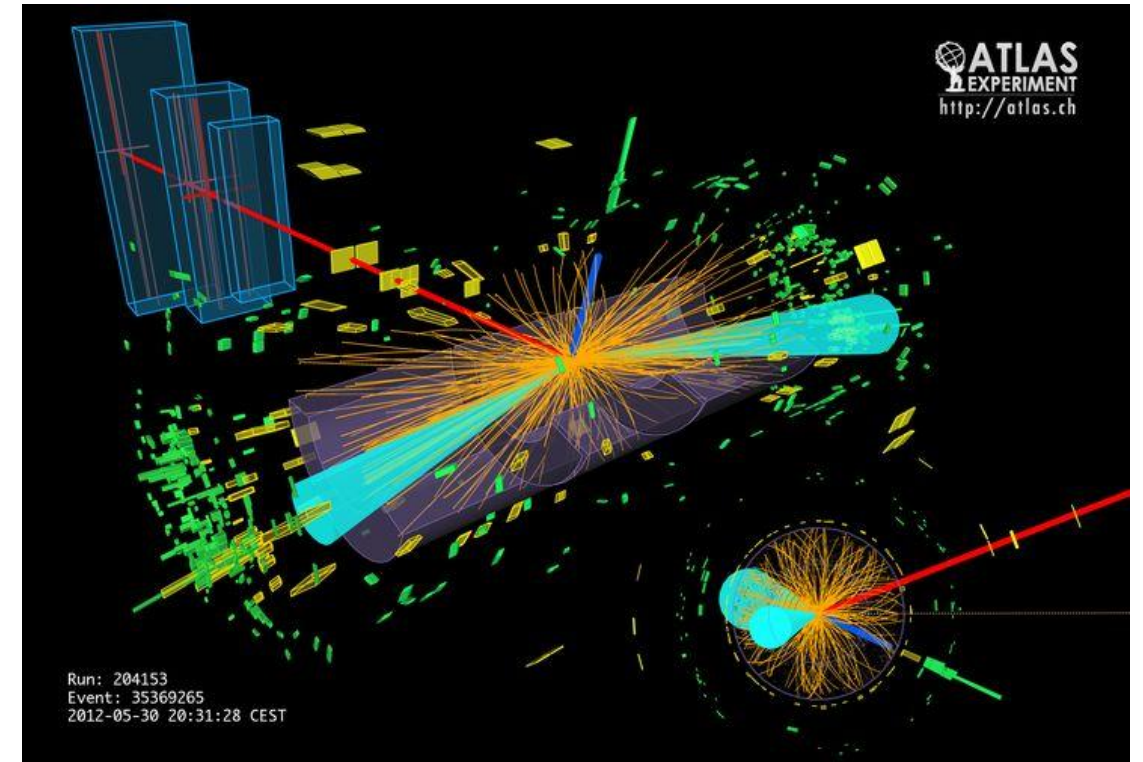
- Schrodinger equation as energy conservation and eigenvalue equation
- Fourier conjugates & uncertainty principle
- Confined vs scattering states
- Common 1D scattering

## 2. S-matrix and its properties

- 1D solutions using S matrix
- Resonance

## 3. 3D potentials and cross section formula

- Partial waves & Phase shifts
- Hard sphere solution
- Bound states & scattering length



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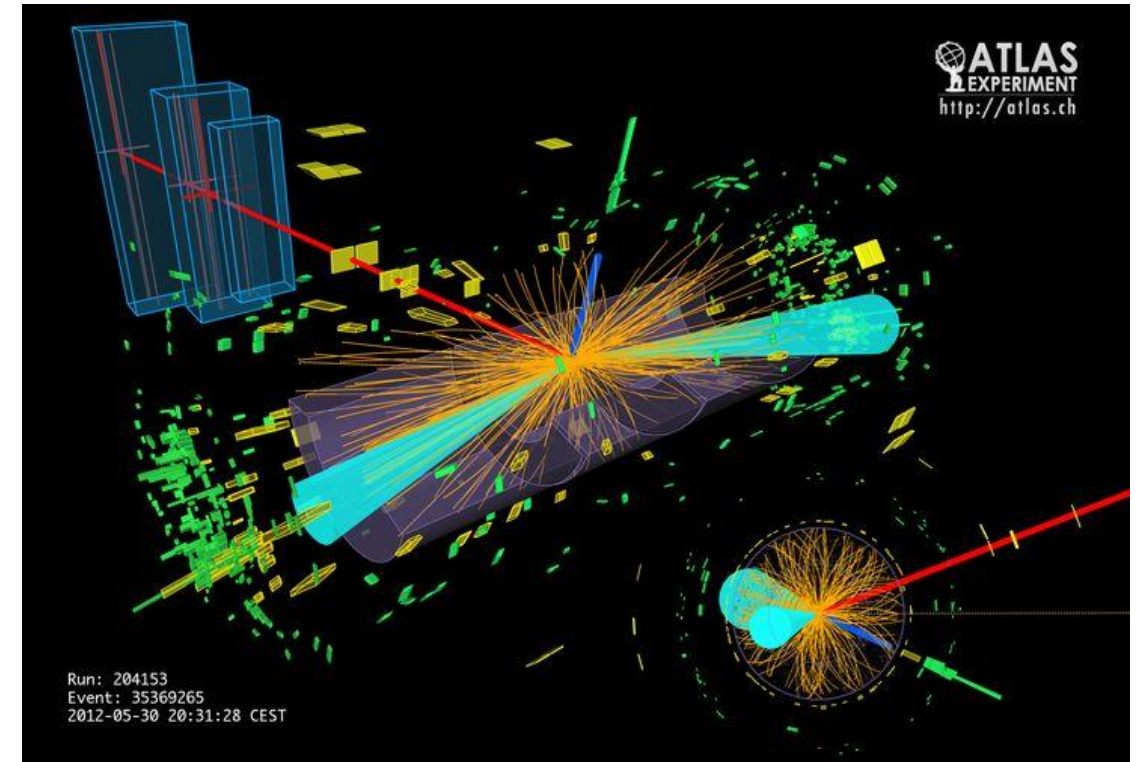
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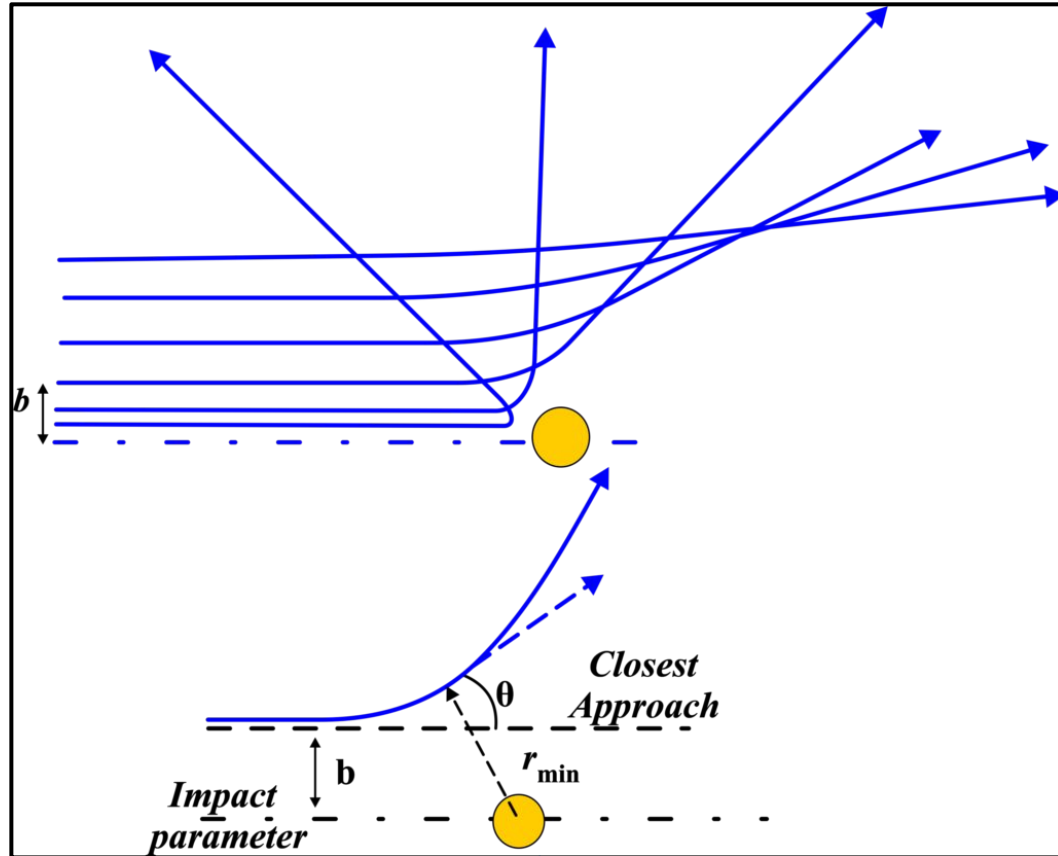
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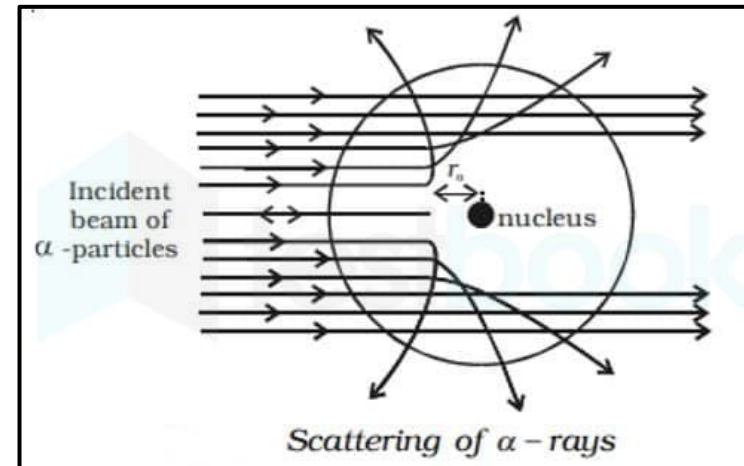
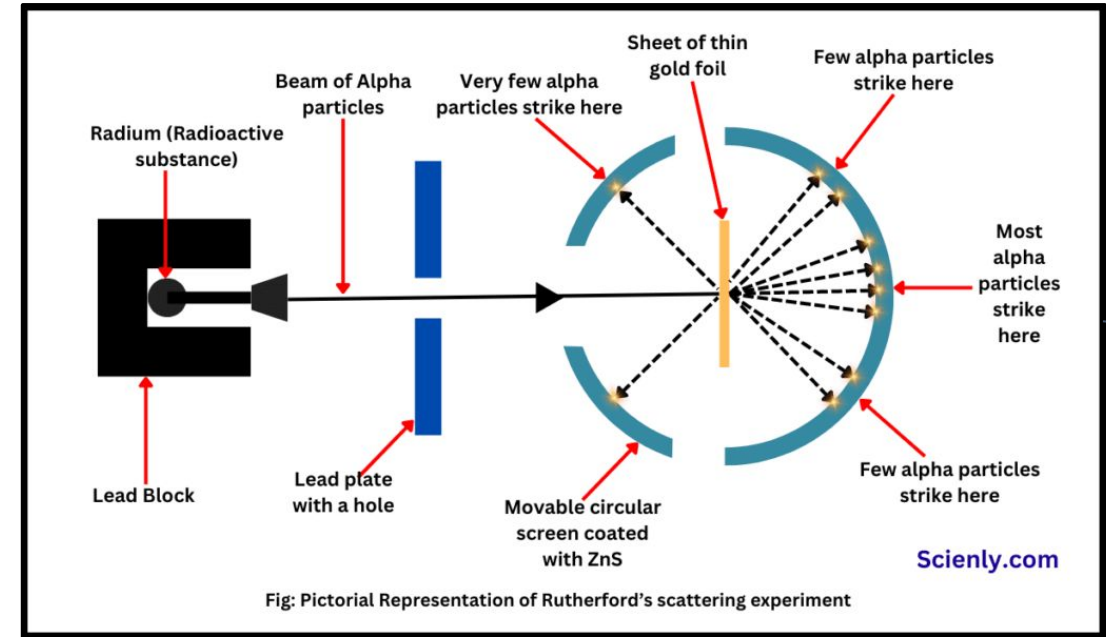
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# What is meant by scattering?



General 3D scattering setup

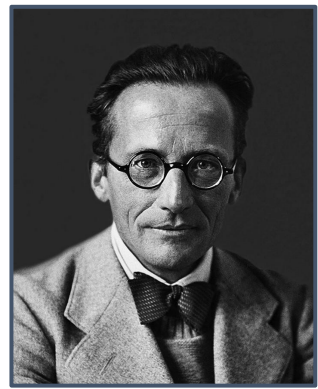


Microscopic view

Rutherford's scattering  
(One of the most famous  
scattering experiments  
ever conducted)



# Schrodinger equation as energy conservation & eigenvalue equation



The Schrödinger equation is nothing mysterious. It is simply the **quantum version of energy conservation**.

In classical physics, we write:

$$E = T + V = \frac{p^2}{2m} + V(x).$$

In quantum mechanics, we keep the same idea, but with a twist: **energy is no longer just a number, it acts on the state of the system.**

So we write:

$$\hat{H} = \hat{T} + \hat{V}, \quad \hat{H}\psi = E\psi.$$

Here,  $\hat{H}$  (the Hamiltonian) operates on the state  $\psi$  and returns the same state scaled by  $E$ . These special states are called **eigenstates**, and the corresponding  $E$  must be **real**, because energy is a physically measurable quantity. This is why  $\hat{H}$  is required to be **Hermitian**.

Think of  $\psi(x, t)$  as the **complete description of the particle**. Not its position, not its velocity, but *everything that can possibly be known about it*. Mathematically, it lives in a **Hilbert space**: a space of well-behaved functions that are continuous, square-integrable, and normalizable.

You can think of it in two equivalent ways:

- as a **continuous function**  $\psi(x, t)$ ,
- or as an abstract **vector**  $|\psi\rangle$  in Hilbert space.

The key point is:  $\psi$  itself is complex-valued and not directly observable. What we *can* observe is:

$$|\psi(x, t)|^2,$$

which gives the **probability density** of finding the particle at position  $x$ .

So in one dimension:

$$|\psi(x, t)|^2 = \text{PDF of position.}$$

This function tells you how the particle is **spread out in space**. The total probability must be 1:

$$\int |\psi(x, t)|^2 dx = 1,$$

which is why  $\psi$  must be **normalizable**. When dealing with multiple states, they are chosen to be **orthonormal**, ensuring consistency of probabilities.

# Fourier conjugates and uncertainty principle

From classical mechanics, Newton's second law

$$F = \frac{dp}{dt}$$

tells us that **momentum changes only when a force acts**. In the absence of force, momentum remains constant, and the particle simply **continues moving through space**. This already hints at a deeper idea: momentum is associated with spatial translation.

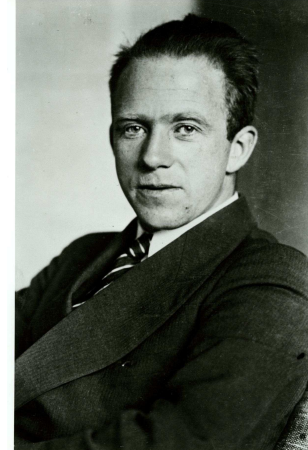
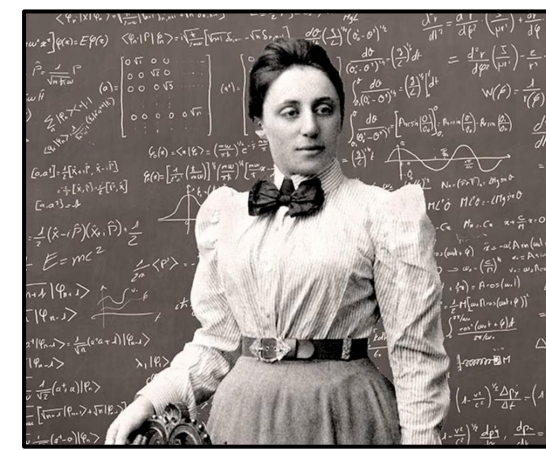
Similarly:

- **momentum** → generates translations in space,
- **energy** → generates translations in time,
- **angular momentum** → generates rotations.

This symmetry is made precise through Noether's theorem and the Poisson bracket structure, and in quantum mechanics leads to the operator forms:

$$\hat{p} = -i\hbar \frac{\partial}{\partial x}, \quad \hat{H} = i\hbar \frac{\partial}{\partial t},$$

with analogous differential forms for angular momentum. The constant  $\hbar$  enters through the de Broglie relation.



Because momentum generates spatial translation, the same quantum state can be described either in position space or in momentum (wave-number) space. These are not different objects, but two representations of the same state, related by Fourier transform:

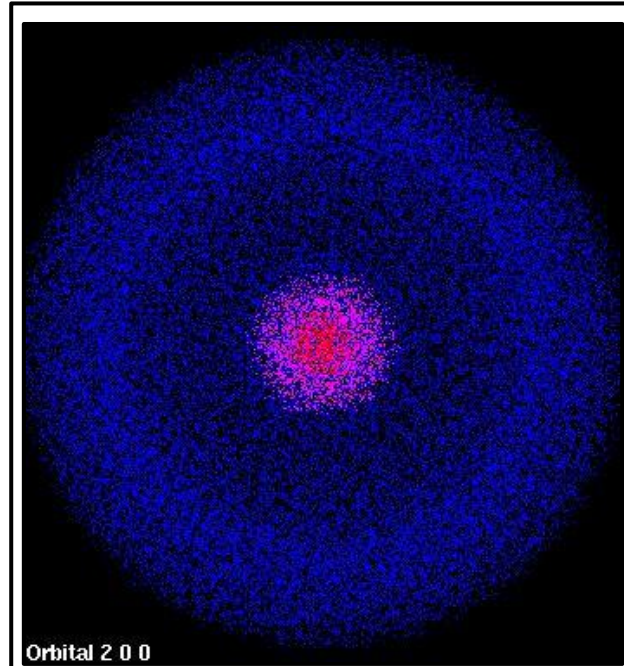
$$\psi(x) = \frac{1}{\sqrt{2\pi}} \int \psi(k) e^{ikx} dk, \quad \psi(k) = \frac{1}{\sqrt{2\pi}} \int \psi(x) e^{-ikx} dx.$$

This relation already encodes the physics. A sharply localized  $\psi(x)$  cannot be built from a single oscillation  $e^{ikx}$ ; it requires a superposition of many different  $k$ -components interfering together. Conversely, if  $\psi(k)$  is sharply peaked at a single value of  $k$  (i.e., well-defined momentum), then the integral reconstructs a long, extended wave in  $x$ , spread out over space.

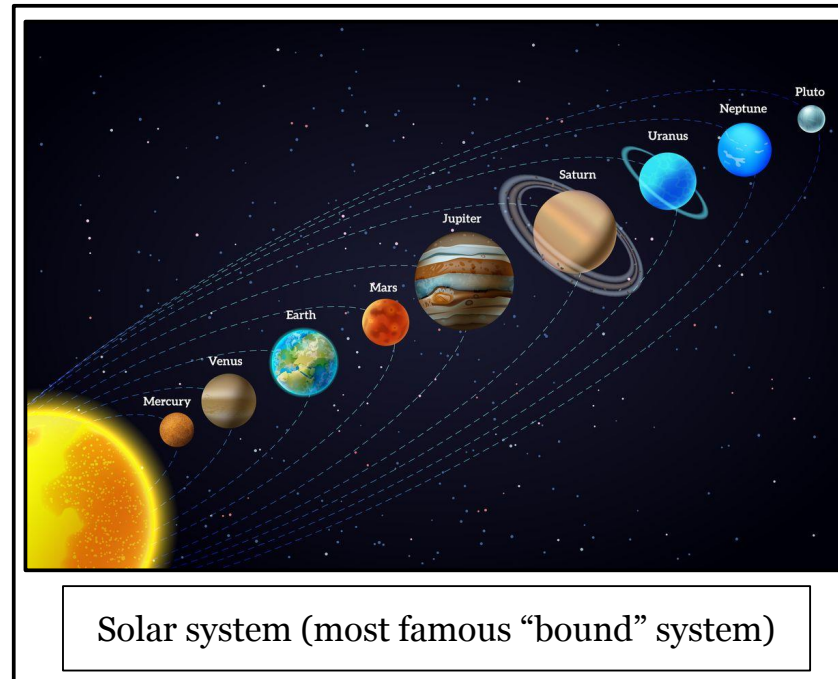
So localization in position is achieved by mixing many momenta, while fixing momentum forces the state to delocalize. The uncertainty principle is nothing more than this Fourier structure expressed quantitatively.

# Confined (bound) vs Scattering (unbound) states

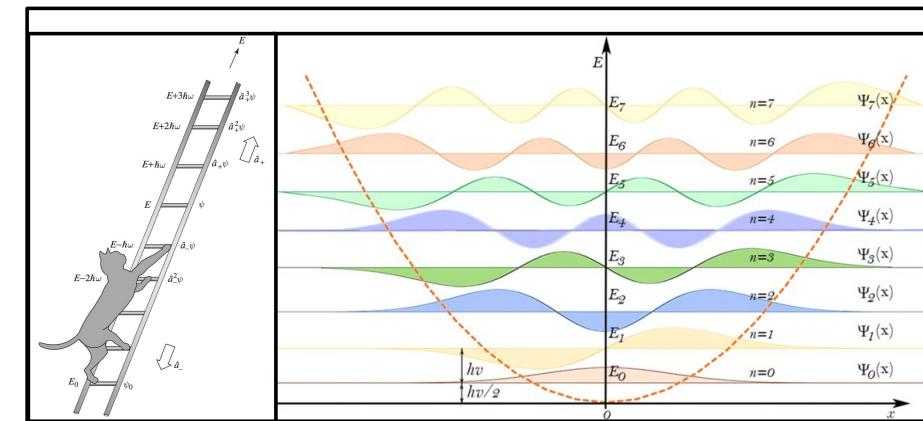
Confinement is the exact opposite of scattering as previously shown.  
Some common examples of “Bound” or “Confined” states.



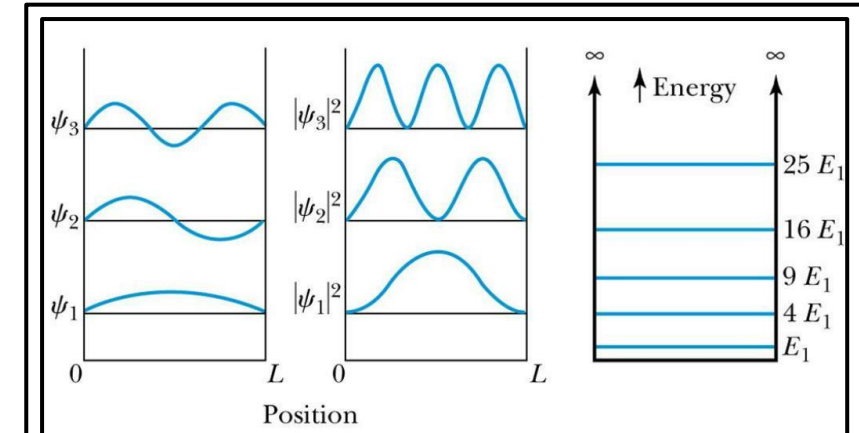
Hydrogen atom (in pic)  
In fact any atomic/  
molecular system



Solar system (most famous “bound” system)



Harmonic oscillator (proportional to  $x^2$ )



Infinite square well potential  
(Rudimentary model for quantum dots)



# Core difference between bound and unbound states.

| Bound state                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Scattering state                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• A bound state arises when the wavefunction is <b>confined</b> within the potential, meaning it remains localized in a finite region of space and decays as we move away from that region.</li><li>• This happens when the energy eigenvalue lies within the bounds set by the potential, typically below its asymptotic value at infinity. Thus, if <math>V(\infty) = 0</math>, <math>E_{\max} &lt; 0</math> always.</li><li>• In this regime, the particle does not have sufficient energy to exist freely outside the potential, so it remains trapped.</li><li>• As a result, only specific standing-wave configurations are allowed, leading to <b>discrete</b> (quantized) energy levels.</li><li>• These states are <b>normalizable</b>, meaning the total probability over all space is finite and equal to one.</li></ul> | <ul style="list-style-type: none"><li>• A scattering state is the opposite regime. Here the potential does not confine the particle globally.</li><li>• Far from the interaction region, the potential effectively vanishes, and the wavefunction behaves like a free particle (<math>V = 0</math> regime). Instead of dying out, it instead acts like a freely propagating wave.</li><li>• Physically, this means the particle comes in with some definite momentum, interacts with the potential over a finite region, and then either reflects or transmits, but it never gets trapped. (<math>E &gt; 0</math>)</li><li>• These states are not normalizable. (<math>E(k) \Rightarrow</math> continuous spectra <math>\equiv \hbar^2 k^2 / 2m \equiv p^2 / 2m \equiv</math> pure kinetic energy)</li><li>• Their energies are not restricted by confinement, so they form a continuous spectrum; positive and spread out over all space.</li></ul> |

The clean distinction is this: **bound states are localized and discrete** because the potential traps them; **scattering states are extended and continuous** because the potential cannot hold them.

# 1D Scattering setup

## Scattering from the Left

We throw the particle in from the left. When it hits the potential, one of two things can happen: it can bounce back, or it can pass straight through. Of course, this being quantum mechanics, it can quite happily do both at the same time. Mathematically, this means that we are looking for a solution which asymptotically takes the form

$$\psi_R(x) \sim \begin{cases} e^{ikx} + re^{-ikx} & x \rightarrow -\infty \\ te^{ikx} & x \rightarrow +\infty \end{cases}$$

$$R = |r|^2 \quad \text{and} \quad T = |t|^2$$

- The **probability current** (probability flux) is

$$\mathbf{j}(\mathbf{r}, t) = \frac{\hbar}{2mi} [\Psi^* (\nabla \Psi) - \Psi (\nabla \Psi^*)].$$

With these definitions the continuity equation reads:

$$\nabla \cdot \mathbf{j} + \frac{\partial \rho}{\partial t} = 0 \Leftrightarrow \nabla \cdot \mathbf{j} + \frac{\partial |\Psi|^2}{\partial t} = 0.$$

Using this probability current as a conserved quantity we get

$$J(x) = -i \frac{\hbar}{2m} \left( \psi^* \frac{d\psi}{dx} - \psi \frac{d\psi^*}{dx} \right)$$

$$\begin{aligned} J(x) &= \frac{\hbar k}{2m} \left[ (e^{-ikx} + r^* e^{+ikx}) (e^{ikx} - r e^{-ikx}) + (e^{ikx} + r e^{-ikx}) (e^{-ikx} - r^* e^{+ikx}) \right] \\ &= \frac{\hbar k}{m} (1 - |r|^2) \quad \text{as } x \rightarrow -\infty \end{aligned}$$

$$J(x) = \frac{\hbar k}{m} |t|^2 \quad \text{as } x \rightarrow +\infty$$

Equating the two gives

$$1 - |r|^2 = |t|^2 \quad \Rightarrow \quad R + T = 1$$

Additionally upon solving the left and right scattering problem respectively and forcing the probability current constraint one shall get:

$$t' = t \quad \text{and} \quad r' = -\frac{r^* t}{t^*}$$

# Common 1D scattering :- “Roadbump” potential

$$V(x) = \begin{cases} 0 & x < -a \\ V_0 & -a \leq x \leq a \\ 0 & x > a \end{cases}$$

In classically allowed regime ( $E > V_0$ ):  
‘ $q$ ’ is real, as a result  $[-a, a]$  is a standing wave solution.

Consider a finite square barrier (“road bump”) of width  $2a$ , centered at the origin:  $V(x) = 0$  for  $x < -a$ ,  $V(x) = V_0$  for  $-a \leq x \leq a$ , and  $V(x) = 0$  for  $x > a$ . A particle is incident from the left.

The time-independent Schrödinger equation can be written in nondimensionalized form as

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E - V(x))\psi = 0$$

Define the wavevectors

$$k = \sqrt{\frac{2mE}{\hbar^2}}, \quad q = \sqrt{\frac{2m(E - V_0)}{\hbar^2}}$$

For the classically allowed case  $E > V_0$ , both  $k$  and  $q$  are real.

The wavefunction ansatz is

$$\psi(x) = \begin{cases} e^{ikx} + Re^{-ikx}, & x < -a \\ Ae^{iqx} + Be^{-iqx}, & -a \leq x \leq a \\ Te^{ikx}, & x > a \end{cases}$$

Continuity of the wavefunction and its first derivative at  $x = -a$  and  $x = +a$  gives

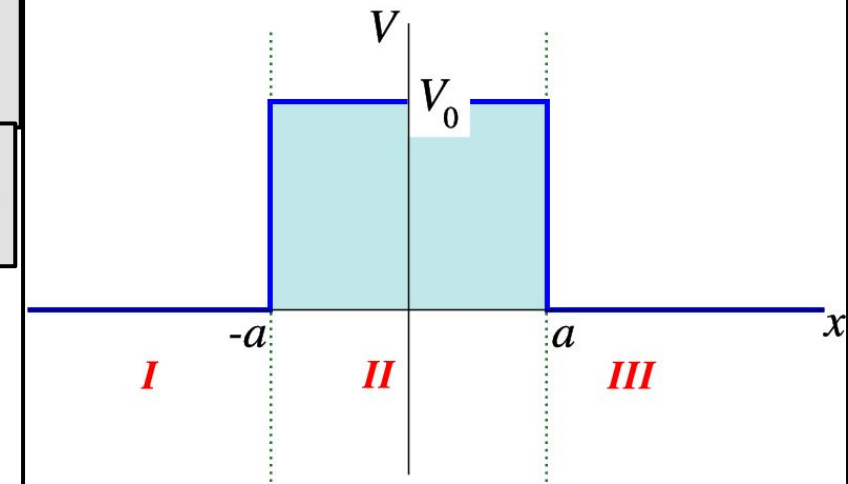
$$\psi_I(-a) = \psi_{II}(-a), \quad \psi_I'(-a) = \psi_{II}'(-a), \quad \psi_{II}(a) = \psi_{III}(a), \quad \psi_{II}'(a) = \psi_{III}'(a)$$

Solving these relations yields the reflection and transmission coefficients

$$R = |R|^2, \quad T = |T|^2 = \frac{1}{1 + \frac{V_0^2}{4E(E - V_0)} \sin^2(2qa)}$$

$$\kappa = \sqrt{\frac{2m(V_0 - E)}{\hbar^2}},$$

$$q = i\kappa$$



In classically “forbidden” regime ( $E < V_0$ ):  
‘ $q$ ’ is imaginary  $\Rightarrow [-a, a]$  is an exponentially decaying function.

This phenomenon is called “quantum tunneling”, because of which the wavefunction might survive only if barrier width is small enough.

Applying the same continuity conditions at  $x = \pm a$  leads to

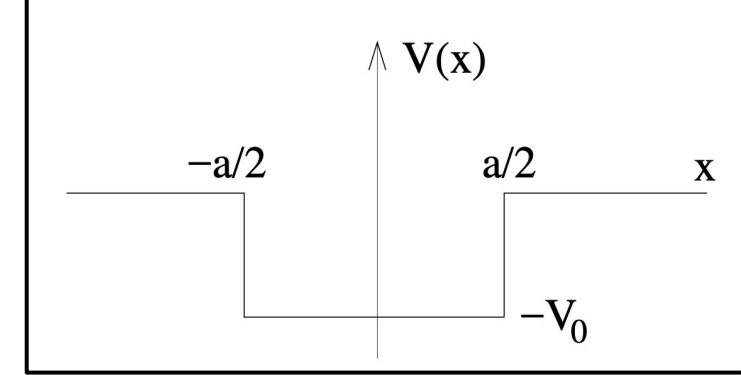
$$T = \frac{1}{1 + \frac{V_0^2}{4E(E - V_0)} \sinh^2(2\kappa a)}$$

In the thick-barrier limit ( $\kappa a \gg 1$ ), this reduces to exponential tunneling  
 $T \sim e^{-4\kappa a}$

The characteristic tunneling decay scale is therefore set by

$$\kappa^{-1} = \frac{\hbar}{\sqrt{2m(V_0 - E)}}$$

# Common 1D scattering :- “Pothole” potential



- The nature of solutions for this system would be very similar to that of the previous system. However, in this case both bound and scattering states exist.
- Wavefunctions with  $E > 0$  would be **scattering** because  $V(\infty) = 0$  and  $E > V(\infty)$ . However the “standing wave” nature for scattered states remains the same.
- For  $E < 0$ , **confinement** occurs and because of the **symmetric** nature of the potential two types of solutions **Even and Odd parity solutions arise**.
- In the potential boundary the wavefunction simply tunnels inside (because  $E < 0$ )

The time-independent Schrödinger equation can be written in nondimensionalized form as

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E - V(x))\psi = 0$$

For scattering states with  $E > 0$ , define the wavevectors

$$k = \sqrt{\frac{2mE}{\hbar^2}}, \quad q = \sqrt{\frac{2m(E + V_0)}{\hbar^2}}$$

The wavefunction ansatz is

$$\psi(x) = \begin{cases} e^{ikx} + Re^{-ikx}, & x < -a \\ [4pt] Ae^{iqx} + Be^{-iqx}, & -a \leq x \leq a \\ [4pt] Te^{ikx}, & x > a \end{cases}$$

Continuity of the wavefunction and its first derivative at  $x = -a$  and  $x = +a$  gives

$$\psi_I(-a) = \psi_{II}(-a), \quad \psi_I'(-a) = \psi_{II}'(-a), \quad \psi_{II}(a) = \psi_{III}(a), \quad \psi_{II}'(a) = \psi_{III}'(a)$$

Solving these matching conditions yields the transmission and reflection coefficients

$$T = |T|^2 = \frac{1}{1 + \frac{V_0^2}{4E(E+V_0)} \sin^2(2qa)}, \quad R = |R|^2 = 1 - T$$

The oscillatory factor  $\sin^2(2qa)$  reflects interference inside the well. At the resonant values  $2qa = n\pi$ , the reflection vanishes and transmission becomes complete.

For odd parity, the wavefunction is

$$\psi(x) = \begin{cases} -Ce^{\kappa x}, & x < -a \\ [4pt] A \sin(qx), & -a \leq x \leq a \\ [4pt] Ce^{-\kappa x}, & x > a \end{cases}$$

Continuity at  $x = a$  gives the corresponding condition

$$\kappa = -q \cot(qa)$$

For even parity, the wavefunction is

$$\psi(x) = \begin{cases} Ce^{\kappa x}, & x < -a \\ [4pt] A \cos(qx), & -a \leq x \leq a \\ [4pt] Ce^{-\kappa x}, & x > a \end{cases}$$

Continuity at  $x = a$  gives the bound-state condition

$$\kappa = q \tan(qa)$$

Together with

$$q^2 + \kappa^2 = \frac{2mV_0}{\hbar^2}$$

- These transcendental equations determine the allowed bound-state energies.
- The even-parity equation always admits at least one solution, so a finite attractive well always supports at least one bound state, while additional even and odd states appear as the well becomes deeper or wider.



# Common 1D scattering :- $\delta$ -function barrier

Consider a delta-function barrier at the origin:  $V(x) = \lambda \delta(x)$ , with  $\lambda > 0$ . A particle is incident from the left.

The time-independent Schrödinger equation in nondimensionalized form is

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} (E - \lambda \delta(x)) \psi = 0$$

Away from  $x = 0$ , the potential vanishes and the equation reduces to the free-particle form. Define the wavevector

$$k = \sqrt{\frac{2mE}{\hbar^2}}$$

Now integrate this equation over a tiny symmetric interval around the origin:

$$\int_{-\epsilon}^{+\epsilon} \left[ -\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + \lambda \delta(x) \psi(x) \right] dx = \int_{-\epsilon}^{+\epsilon} E \psi(x) dx$$

The wavefunction is continuous at  $x = 0$ , but its derivative has a discontinuity obtained by integrating the Schrödinger equation across the origin. The matching conditions are

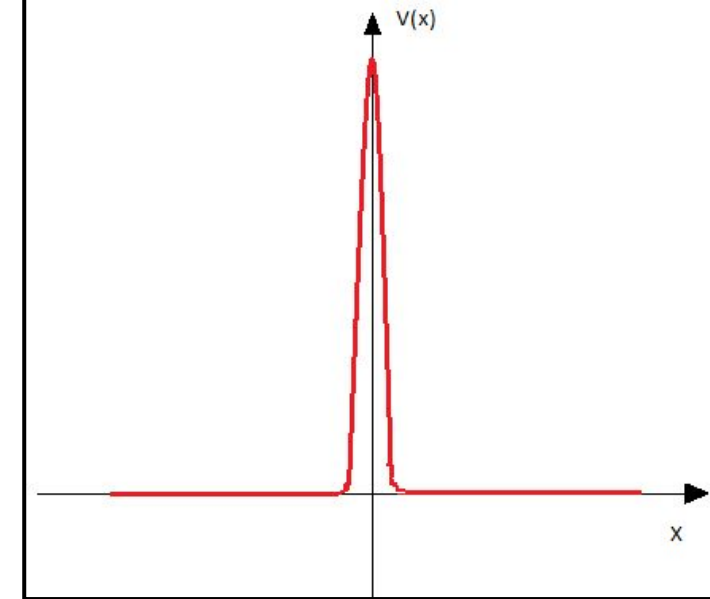
$$\psi(0^-) = \psi(0^+), \quad \psi(0^+) - \psi(0^-) = \frac{2m\lambda}{\hbar^2} \psi(0)$$

Solving these relations gives the reflection and transmission coefficients

$$R = |R|^2 = \frac{\gamma^2}{k^2 + \gamma^2}, \quad T = |T|^2 = \frac{k^2}{k^2 + \gamma^2}, \quad \text{where } \gamma = \frac{m\lambda}{\hbar^2}$$

The wavefunction ansatz is

$$\psi(x) = \begin{cases} e^{ikx} + Re^{-ikx}, & x < 0 \\ [4pt] Te^{ikx}, & x > 0 \end{cases}$$



For an attractive delta potential  $V(x) = -\lambda \delta(x)$ , with  $\lambda > 0$ , bound states exist with energy  $E < 0$ . Define

$$\kappa = \sqrt{\frac{-2mE}{\hbar^2}}$$

The bound-state wavefunction decays exponentially on both sides of the origin and is even:

$$\psi(x) = Ae^{-\kappa|x|}$$

The derivative discontinuity condition gives

$$\psi(0^+) - \psi(0^-) = -\frac{2m\lambda}{\hbar^2} \psi(0)$$

which yields the bound-state condition

$$\kappa = \frac{m\lambda}{\hbar^2}$$

Therefore only single energy eigenstate exists with energy:

$$E = -\frac{m\lambda^2}{2\hbar^2}$$

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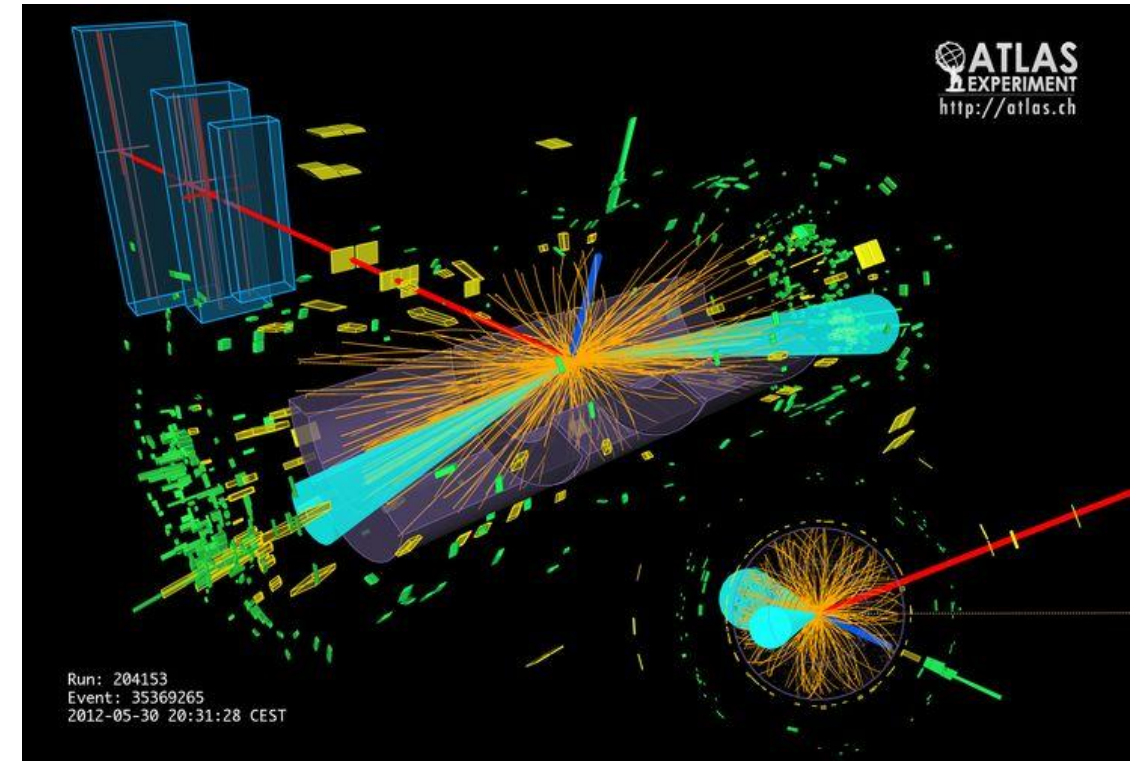
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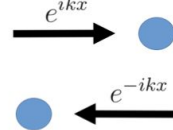
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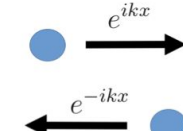


# S-matrix and its properties : unitarity & parity basis

The *S-matrix* is a convenient way of packaging the information about reflection and transmission coefficients. It is useful both because it highlights new features of the problem, and because it generalises to scattering in higher dimensions.

$$\text{Ingoing} \begin{cases} \text{right-moving:} & \mathcal{I}_R(x) = e^{+ikx} & x \rightarrow -\infty \\ \text{left-moving:} & \mathcal{I}_L(x) = e^{-ikx} & x \rightarrow +\infty \end{cases}$$


Similarly, there are two outgoing asymptotic wavefunctions,

$$\text{Outgoing} \begin{cases} \text{right-moving:} & \mathcal{O}_R(x) = e^{+ikx} & x \rightarrow +\infty \\ \text{left-moving:} & \mathcal{O}_L(x) = e^{-ikx} & x \rightarrow -\infty \end{cases}$$


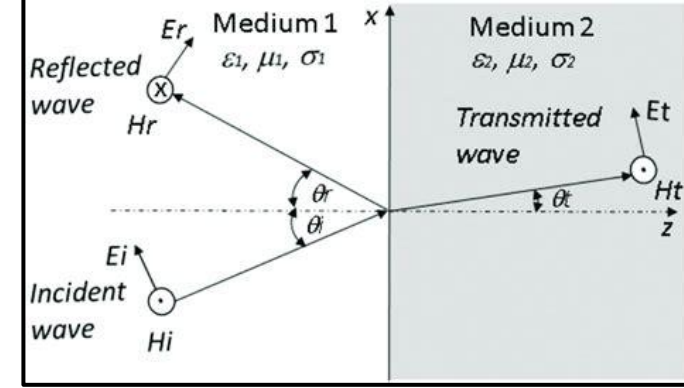
$$\begin{pmatrix} \psi_R \\ \psi_L \end{pmatrix} = \begin{pmatrix} \mathcal{I}_R \\ \mathcal{I}_L \end{pmatrix} + \mathcal{S} \begin{pmatrix} \mathcal{O}_R \\ \mathcal{O}_L \end{pmatrix} \longleftrightarrow \mathcal{S} = \begin{pmatrix} t & r \\ r' & t' \end{pmatrix}$$

Property of Unitarity :

$$\mathcal{S}\mathcal{S}^\dagger = \begin{pmatrix} |t|^2 + |r|^2 & tr'^* + rt'^* \\ t^*r' + t'r^* & |t'|^2 + |r'|^2 \end{pmatrix}$$

The off-diagonal elements vanish by virtue of the relations  $t' = t$  and  $r' = -r^*t/t^*$  (from probability current proof).

Meanwhile, the diagonal elements are equal to 1 (again conservation of probability current).



For symmetric potential i.e  $V(x) = V(-x)$

The Hamiltonian commutes with the parity operator  $P: f(x) \rightarrow f(-x)$ , hence eigenstates can be chosen with definite parity. Since  $P^2 = 1$ , the eigenvalues are  $\lambda = \pm 1$ , giving even and odd sectors:

$$\psi_+(x) = \psi_+(-x), \quad \psi_-(x) = -\psi_-(-x)$$

The usual scattering states  $\psi_R, \psi_L$  (incoming from right and left with asymptotic plane waves  $e^{\pm ikx}$ ) are not parity eigenstates but satisfy the symmetry relation  $\psi_L(x) = \psi_R(-x)$

The eigenstates in the parity basis

$$\psi_+(x) = \psi_R(x) + \psi_L(x), \quad \psi_-(x) = -\psi_R(x) + \psi_L(x)$$

which obey

$$P\psi_\pm = \pm\psi_\pm$$

# Parity Basis (diagonal trick) & phase shift

These are related to our earlier modes by a simple change of basis,

$$\begin{pmatrix} \mathcal{I}_+ \\ \mathcal{I}_- \end{pmatrix} = \mathcal{M} \begin{pmatrix} \mathcal{I}_R \\ \mathcal{I}_L \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} \mathcal{O}_+ \\ \mathcal{O}_- \end{pmatrix} = \mathcal{M} \begin{pmatrix} \mathcal{O}_R \\ \mathcal{O}_L \end{pmatrix} \quad \text{with} \quad \mathcal{M} = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$$

We can define an S-matrix with respect to this parity basis. In analogy with (6.9), we write asymptotic solutions as

$$\begin{pmatrix} \psi_+ \\ \psi_- \end{pmatrix} = \begin{pmatrix} \mathcal{I}_+ \\ \mathcal{I}_- \end{pmatrix} + \mathcal{S}^P \begin{pmatrix} \mathcal{O}_+ \\ \mathcal{O}_- \end{pmatrix}$$

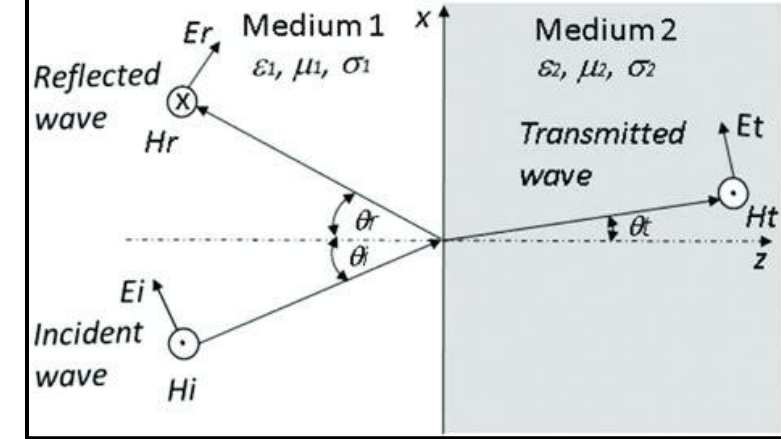
where we use the notation  $\mathcal{S}^P$  to denote the S-matrix with respect to the parity basis.

We write

$$\mathcal{S}^P = \begin{pmatrix} S_{++} & S_{+-} \\ S_{-+} & S_{--} \end{pmatrix}$$

This is related to our earlier S-matrix by a change of basis. We have

$$\boxed{\mathcal{S}^P = \mathcal{M} S \mathcal{M}^{-1}} = \begin{pmatrix} t + (r + r')/2 & (r - r')/2 \\ (r' - r)/2 & t - (r + r')/2 \end{pmatrix}$$



As you may expect, this basis is particularly useful if the underlying potential is symmetric, so  $V(x) = V(-x)$ . In this case we have  $r = r'$  and the S-matrix becomes diagonal. The diagonal components are simply

$$S_{++} = t + r \quad \text{and} \quad S_{--} = t - r$$

In fact, because  $\mathcal{S}^P$  is unitary, each of these components must be a phase. This follows because  $r$  and  $t$  are not independent. First, they obey  $|r|^2 + |t|^2 = 1$ . Moreover, when  $r' = r$ , the relation (6.6) becomes

$$\boxed{rt^* + r^*t = 0} \Rightarrow \text{Re}(rt^*) = 0$$

This is enough to ensure that both  $S_{++}$  and  $S_{--}$  are indeed phases. We write them as

$$S_{++} = e^{2i\delta_+(k)} \quad \text{and} \quad S_{--} = e^{2i\delta_-(k)}$$

We learn that for scattering off a symmetric potential, all the information is encoded in two momentum-dependent *phase shifts*,  $\delta_{\pm}(k)$  which tell us how the phases of the outgoing waves  $\mathcal{O}_{\pm}$  are changed with respect to the ingoing waves  $\mathcal{I}_{\pm}$ .



# 1D solution using s matrix : pothole revisited

For the even-parity channel, the exterior and interior solutions may be written as

$$\psi_+(x) = \begin{cases} e^{-ik|x|} + S_{++} e^{ik|x|}, & |x| > a/2 \\ [6pt] A (e^{iqx} + e^{-iqx}), & |x| < a/2 \end{cases}$$

so that the interior wavefunction is explicitly even. Evaluating at  $x=a/2$ , continuity of  $\Psi'/\Psi$  (**Robin type boundary condition**), gives

$$\frac{-ik e^{-ika/2} + ik S_{++} e^{ika/2}}{e^{-ika/2} + S_{++} e^{ika/2}} = \frac{iq e^{iqa/2} - iq e^{-iqa/2}}{e^{iqa/2} + e^{-iqa/2}} = -q \tan\left(\frac{qa}{2}\right)$$

and solving for  $S_{++}$  yields

$$S_{++} = -e^{-ika} \frac{q \tan(qa/2) - ik}{q \tan(qa/2) + ik}$$

Similarly for odd parity states, robin boundary condition at  $x = a/2$  gives

$$\frac{-ik e^{-ika/2} + ik S_{--} e^{ika/2}}{e^{-ika/2} - S_{--} e^{ika/2}} = \frac{iq e^{iqa/2} + iq e^{-iqa/2}}{e^{iqa/2} - e^{-iqa/2}} = q \cot\left(\frac{qa}{2}\right)$$

and solving for  $S_{--}$  gives

$$S_{--} = e^{-ika} \frac{q + ik \tan(qa/2)}{q - ik \tan(qa/2)}$$

Method of solving :

1. We start with the 'roadbump' potential and assume standing waves inside the interaction region (classically allowed regime aka NO tunneling).
2. We solve using the S-matrix formalism at  $x= a/2$  boundary for first diagonal (even states) and then second diagonal (odd states).

- Now let us observe both observe the complex poles of  $S_{++}$  and  $S_{--}$ .
- For bound states,  $k = i\lambda$ , subsequently this produces the transcendental equations like before.
- But this time we solved it using S-matrix formulation!

# OUTLINE

## 1. Introduction

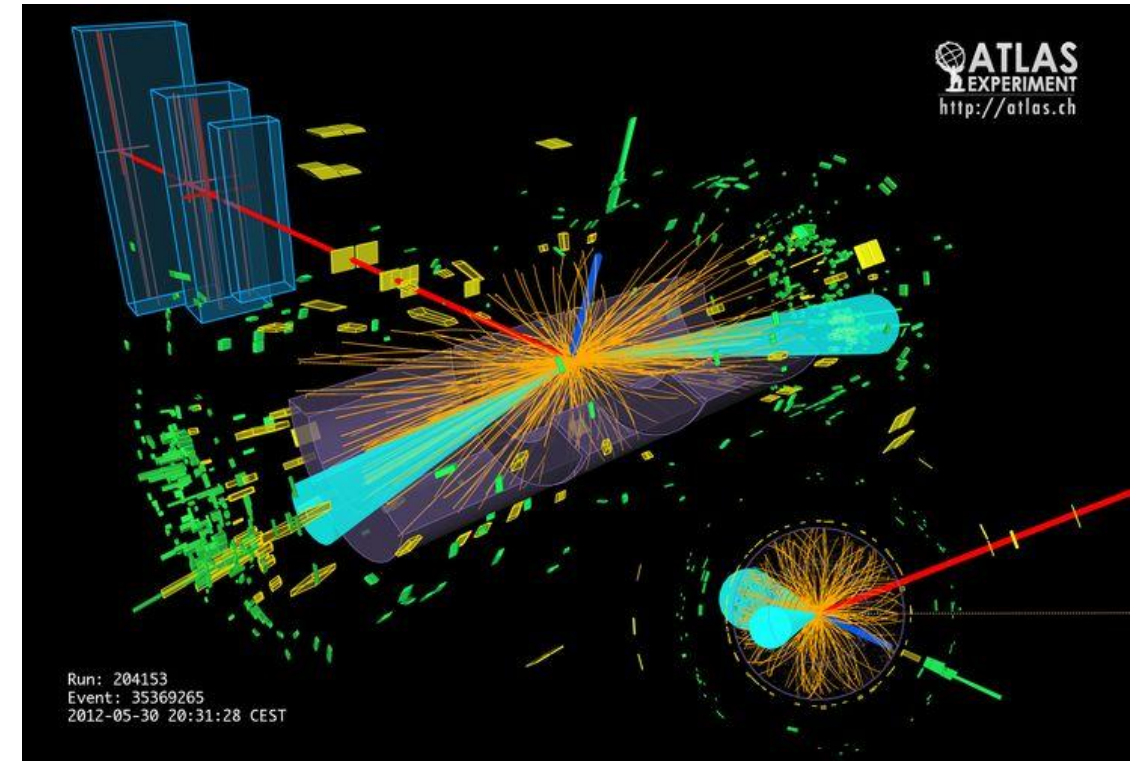
- Schrodinger equation as energy conservation and eigenvalue equation
- Fourier conjugates & uncertainty principle
- Confined vs scattering states
- Common 1D scattering

## 2. S-matrix and its properties

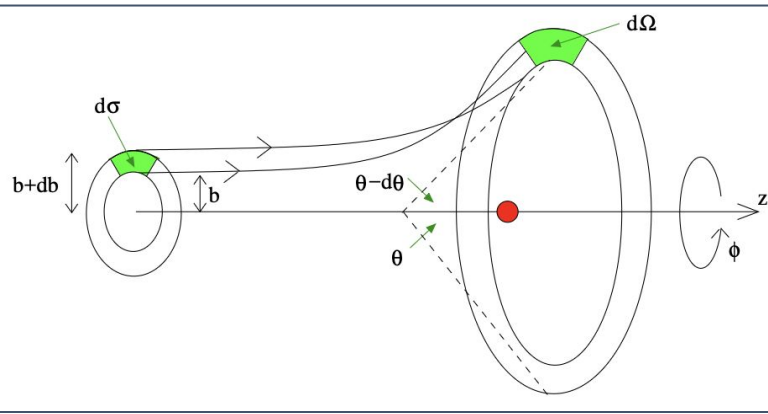
- 1D solutions using S matrix
- Resonances

## 3. 3D potentials and cross section formula

- Partial waves & Phase shifts
- Hard sphere solution
- Bound states & scattering length



# 3D potentials and cross section formula : cross section



$$d\sigma = b \, d\phi \, db$$

The particles within  $d\sigma$  will evolve to lie in a cone of solid angle  $d\Omega$ , given by

$$d\Omega = \sin \theta \, d\phi \, d\theta$$

where, for central potentials, the infinitesimal angles  $d\phi$  are the same in both these formulae. The *differential cross-section* is defined to be

$$\frac{d\sigma}{d\Omega} = \frac{b}{\sin \theta} \left| \frac{db}{d\theta} \right|$$

Cross section

$$\frac{d\sigma}{d\Omega} = \frac{\text{Scattered flux}}{\text{Incident flux}}$$

3D wavefunction => plane wave incident, spherical wave scattered

$$\psi(\mathbf{r}) \sim e^{ikz} + f(\theta, \phi) \frac{e^{ikr}}{r}$$

For the scattered part

$$\psi_{\text{sc}}(\mathbf{r}) = f(\theta) \frac{e^{ikr}}{r}$$

the large- $r$  gradient is dominated by the radial derivative, so that

$$\nabla \psi_{\text{sc}} \sim ik f(\theta) \frac{e^{ikr}}{r} \hat{\mathbf{r}}$$

and the corresponding current becomes purely radial,

$$\mathbf{J}_{\text{sc}} = \frac{\hbar k}{m} \frac{|f(\theta)|^2}{r^2} \hat{\mathbf{r}}$$

The flux of scattered particles through an area element on a sphere of radius  $r$  is therefore

$$\mathbf{J}_{\text{sc}} \cdot \hat{\mathbf{r}} \, dA = \frac{\hbar k}{m} \frac{|f(\theta)|^2}{r^2} (r^2 d\Omega) = \frac{\hbar k}{m} |f(\theta)|^2 d\Omega$$

The differential cross section is defined as the scattered flux into the solid angle  $d\Omega$ , divided by the incident flux per unit area. Since the incident flux has magnitude

$\hbar k/m$ , this gives

$$\frac{d\sigma}{d\Omega} = \frac{(\hbar k/m) |f(\theta)|^2 d\Omega}{\hbar k/m}$$

and hence the standard result

$$\frac{d\sigma}{d\Omega} = |f(\theta)|^2$$

# Phase shifts

- In three-dimensional spherical geometry, the time-independent Schrödinger equation for a central potential  $V(r)$  separates as  $\psi(\mathbf{r}) = R_l(r) Y_l^m(\theta, \phi)$ , where  $Y_l^m(\theta, \phi)$  are spherical harmonics (angular eigenfunctions) and the radial function satisfies  $\frac{d^2 u_l}{dr^2} + \left[ k^2 - \frac{l(l+1)}{r^2} - \frac{2m}{\hbar^2} V(r) \right] u_l = 0$ ,  $u_l = rR_l$
- The term  $\frac{l(l+1)}{r^2}$  acts as an effective centrifugal barrier, forcing each angular momentum channel  $l$  to behave like an independent 1D scattering problem in  $r$
- In the asymptotic region  $r \rightarrow \infty$  where  $V(r) \rightarrow 0$ , the radial equation reduces to  $\frac{d^2 u_l}{dr^2} + \left[ k^2 - \frac{l(l+1)}{r^2} \right] u_l = 0$ , whose solutions are spherical Bessel-type functions
- The general radial solution is a linear combination of  $j_l(kr)$  (regular at origin) and  $n_l(kr)$  (singular at origin), or equivalently spherical Hankel functions  $h_l^{(1)}(kr) \sim \frac{e^{ikr}}{kr}$  and  $h_l^{(2)}(kr) \sim \frac{e^{-ikr}}{kr}$
- Asymptotically, the spherical Bessel functions behave as  $j_l(kr) \sim \frac{1}{kr} \sin\left(kr - \frac{l\pi}{2}\right)$  and  $n_l(kr) \sim -\frac{1}{kr} \cos\left(kr - \frac{l\pi}{2}\right)$ , so every partial wave naturally carries a built-in phase shift of  $-l\pi/2$  arising purely from angular momentum
- The most general asymptotic radial solution for each  $l$  is therefore  $u_l(r) \sim \sin\left(kr - \frac{l\pi}{2} + \delta_l\right)$ , where  $\delta_l$  is the scattering phase shift encoding the effect of the potential



# Partial waves

- The incident plane wave propagating along  $z$  is  $e^{ikz} = e^{ikr \cos \theta}$  and can be expanded into partial waves as  $e^{ikz} = \sum_{l=0}^{\infty} (2l+1) i^l j_l(kr) P_l(\cos \theta)$
- This expansion shows that even a simple plane wave is actually a superposition of infinitely many angular momentum channels, each weighted by Legendre polynomials  $P_l(\cos \theta)$
- The scattered wave is purely outgoing and has the universal asymptotic form  $\psi_{\text{sc}}(\mathbf{r}) \sim f(\theta) \frac{e^{ikr}}{r}$ , which corresponds to spherical waves propagating radially outward
- The full wavefunction at large  $r$  is written as  $\psi(\mathbf{r}) \sim e^{ikz} + f(\theta) \frac{e^{ikr}}{r}$
- Rewriting both the plane wave and scattered wave in the same partial wave basis leads to  $\psi(\mathbf{r}) \sim \sum_{l=0}^{\infty} (2l+1) P_l(\cos \theta) \frac{1}{2ikr} [e^{-ikr} - (-1)^l e^{ikr} + 2if_l e^{ikr}]$
- Comparing with the general asymptotic form of  $u_l(r)$ , one identifies  $f_l = \frac{e^{2i\delta_l} - 1}{2i}$  and hence the full scattering amplitude becomes  $f(\theta) = \sum_{l=0}^{\infty} (2l+1) \frac{e^{2i\delta_l} - 1}{2ik} P_l(\cos \theta)$

- Physically each  $l$  is an independent scattering channel, the potential modifies only the phase.
- No change in amplitude occurs, only a phase shift, so interference between incoming and outgoing waves produces scattering.

# Recap so far before we solve for specific potentials

For the same impact parameter we have orbital angular momentum 'l' dependent channel separation and the scattering amplitude f(θ) given by :

$$f(\theta) = \frac{1}{2ik} \sum_{l=0}^{\infty} (2l+1) (e^{2i\delta_l} - 1) P_l(\cos \theta)$$

Partial wave was expanded for the initial incident plane wave using  $z = r \cos(\theta)$  given by :

$$e^{ikz} = \sum_{l=0}^{\infty} (2l+1) i^l j_l(kr) P_l(\cos \theta)$$

and using bessel function asymptotics it becomes

$$e^{ikz} \sim \sum_{l=0}^{\infty} (2l+1) P_l(\cos \theta) \frac{1}{2ikr} [e^{ikr} - (-1)^l e^{-ikr}]$$

Total wavefunction should be :

$$\sum_{l=0}^{\infty} (2l+1) P_l(\cos \theta) \frac{1}{2ikr} [\text{incoming} + \text{outgoing}]$$

Therefore if we modify the f(θ) expansion for the spherical wave:

$$\frac{e^{ikr}}{r} f(\theta) = \sum_l (2l+1) \frac{f_l}{r} P_l(\cos \theta) e^{ikr}$$

We get the general form of our complete wavefunction:

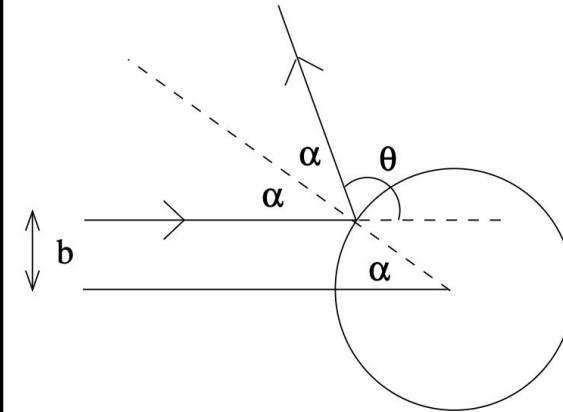
$$\psi(\mathbf{r}) \sim \sum_{l=0}^{\infty} (2l+1) P_l(\cos \theta) \frac{1}{2ikr} [(-1)^{l+1} e^{-ikr} + (1 + 2i f_l) e^{ikr}]$$

We now have:

1. **Partial wave expansion** of both the **incident plane** and **scattered spherical** wave.
2. We also have the partial wave expansion of '**l**'-**dependent channel based scattering amplitude** f(θ).
3. We have **impact parameter** 'b' dependent scattering angle θ formula  $[db/d\theta = (\sin(\theta)/b) (d\sigma/d\Omega)]$ .
4. Hence our **scattering mathematical arsenal** is **complete** and we can now **solve for specific potentials**.
5. Naturally the **hard sphere** ( $V[r < r_0] = \infty$ ) becomes the easiest to solve at first and then we can soften this boundary for attractive ( $V[r < r_0] < 0$ ) & repulsive ( $V(r < r_0) > 0$ ) potentials!

# Solution for the hard sphere

- For a hard sphere of radius  $a$ , the potential is  $V(r) = \infty$  for  $r < a$  and  $V(r) = 0$  for  $r > a$ . The physical content of the infinite barrier is the boundary condition  $\psi(r=a) = 0$ , or equivalently  $R_l(a) = 0$  in each partial-wave channel. Because the potential is spherically symmetric, the angular dependence is completely handled by the spherical harmonics  $Y_l^m(\theta, \phi)$ , which remain unaffected; all nontrivial physics of scattering is therefore encoded in the radial part. In the exterior region, separation of variables reduces the problem to the radial equation whose general solution is of Bessel type, explicitly  $R_l(r) = A j_l(kr) + B n_l(kr)$ , where  $j_l$  and  $n_l$  are the spherical Bessel and spherical Neumann functions
- In the exterior region  $r > a$ , the radial equation is the free radial Schrödinger equation, so the  $l$ -th partial wave is written as a linear combination of the regular and irregular spherical solutions,  $R_l(r) = A_l [\cos \delta_l j_l(kr) - \sin \delta_l n_l(kr)]$ . This is the standard form in which the phase shift  $\delta_l$  is already isolated as the relative coefficient between the spherical Bessel and spherical Neumann pieces
- The reason for this particular combination is that, asymptotically,  $j_l$  and  $n_l$  combine into the shifted oscillatory form associated with scattering, so  $\delta_l$  appears directly as the phase shift of the  $l$ -th partial wave. No further asymptotic derivation is needed here; it is enough to use the known free-space basis  $\{j_l, n_l\}$



# Solution for the hard sphere

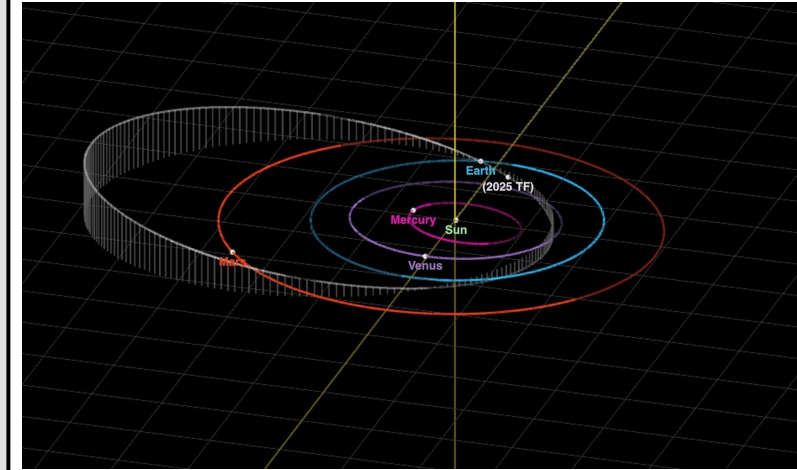
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- Imposing the hard-sphere boundary condition at the surface  $r = a$  gives  $R_l(a) = 0$ , hence  $A_l [\cos \delta_l j_l(ka) - \sin \delta_l n_l(ka)] = 0$ . Since the overall normalization  $A_l$  is nonzero and physically irrelevant, the required condition is  $\cos \delta_l j_l(ka) - \sin \delta_l n_l(ka) = 0$
- Rearranging immediately yields  $\cos \delta_l j_l(ka) = \sin \delta_l n_l(ka)$ , and therefore the hard-sphere phase shift is determined by  $\tan \delta_l = \frac{j_l(ka)}{n_l(ka)}$
- Thus, for hard-sphere scattering, the entire effect of the impenetrable core is encoded in the requirement that the exterior free radial wave vanish at  $r = a$ , and this fixes the phase shift in each angular momentum channel through the ratio of the spherical Bessel and spherical Neumann functions evaluated at the surface



# Bound states (for attractive potential)

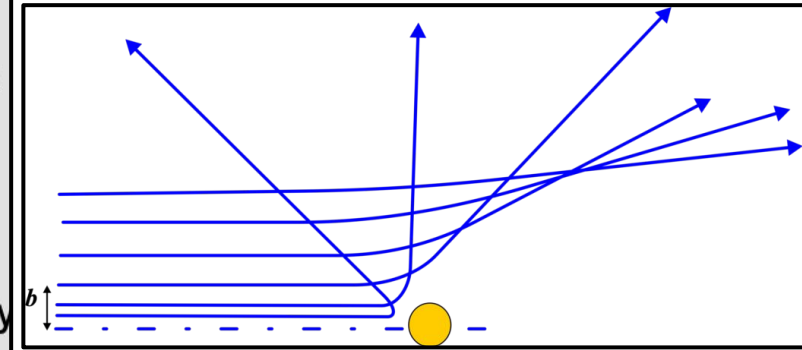
- Consider a short-range attractive potential whose depth is parameterized as  $V_0 = \frac{\hbar^2 \gamma^2}{2m}$ ; bound states correspond to negative energy solutions  $E = -\frac{\hbar^2 \kappa^2}{2m}$  with  $\kappa > 0$ , so the wavefunction must decay at infinity
- The Schrödinger equation separates as  $\psi(\mathbf{r}) = R_l(r)Y_l^m(\theta, \phi)$ , and as before all nontrivial physics sits in the radial equation  $\frac{d^2 u_l}{dr^2} + \left[-\kappa^2 - \frac{l(l+1)}{r^2} + \frac{2m}{\hbar^2} V(r)\right] u_l = 0$ , with  $u_l = rR_l$ ; angular structure is unchanged because the potential is central
- Outside the potential region, where  $V(r) = 0$ , the radial equation becomes  $\frac{d^2 u_l}{dr^2} - \left[\kappa^2 + \frac{l(l+1)}{r^2}\right] u_l = 0$ , whose solutions are modified spherical Bessel functions; asymptotically the physically allowed solution behaves as  $u_l(r) \sim e^{-\kappa r}$ , since normalizability forbids growing exponentials
- Inside the interaction region, the energy is effectively shifted, so the equation resembles a free-particle equation with wave number  $\sqrt{\gamma^2 - \kappa^2}$ ; thus the interior solution is oscillatory and can be written in terms of spherical Bessel functions  $j_l(\sqrt{\gamma^2 - \kappa^2} r)$ , while the irregular solution is discarded by regularity at the origin
- The full solution is therefore piecewise: oscillatory (Bessel-type) inside, exponentially decaying outside; the "general solution" form arises purely from solving the radial equation in these two regions with the appropriate physical constraints (regularity at  $r = 0$ , decay at  $r \rightarrow \infty$ )



Asteroids coming from outer space asymptotically far from the attractive field of a star/planet attain bound state orbits!

# Bound states and Scattering states

- Matching conditions at the boundary (continuity of  $u_l$  and its derivative) quantize the allowed  $\kappa$ , giving discrete bound state energies; physically this is identical to standing-wave quantization but now with exponential tails outside
- The **connection to scattering** comes from analytic continuation: replace  $k \rightarrow i\kappa$  in the scattering solutions, so the asymptotic oscillatory form  $u_l(r) \sim \sin(kr - \frac{l\pi}{2} + \delta_l)$  becomes an exponential decay
- The phase shift  $\delta_l(k)$  encodes how **the scattering solution** deviates from the free solution; bound states appear as special points where the scattering amplitude develops poles, which corresponds to  $\delta_l$  passing through  $\frac{\pi}{2}$  modulo  $\pi$
- More precisely, the phase shift is determined by matching the interior Bessel-type solution to the exterior solution, leading to a relation of the form  $\tan \delta_l = (\text{ratio of interior and exterior solutions at the boundary})$ , structurally identical to the hard-sphere case but now involving modified wave numbers
- Bound states correspond to the condition that the denominator of the scattering amplitude vanishes, which is equivalent to  $\tan \delta_l \rightarrow \infty$ , i.e.  $\delta_l = \frac{\pi}{2} \pmod{\pi}$ ; this is the precise mathematical signal that a normalizable state has formed



1. Repulsion by scattering potential, channel separation due to  $db/d\theta = K(d\sigma/d\Omega)$ .
2. For same  $b$ , 'l' dependent channel separation given by  $f(\theta)$  amplitude partial wave expansion!

KEY :

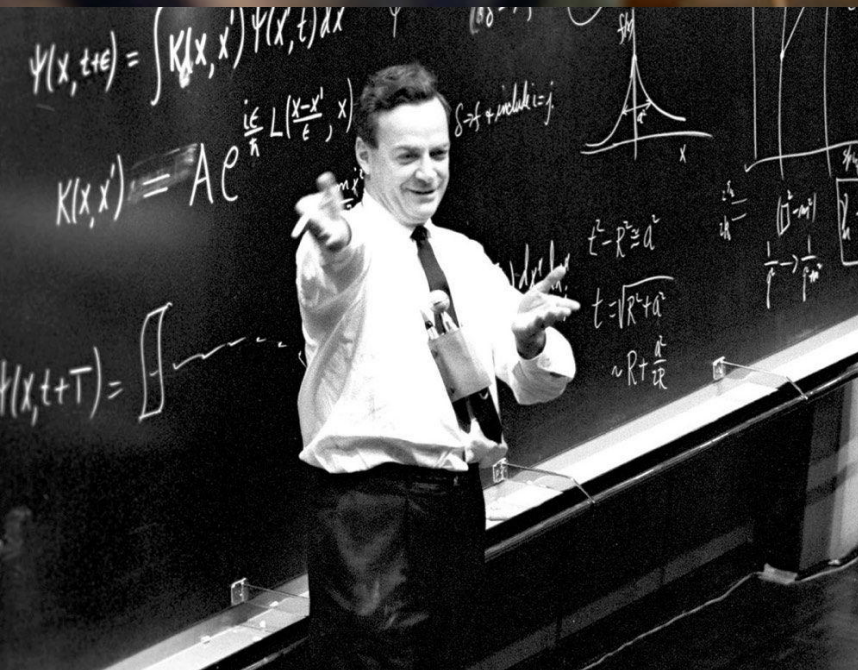
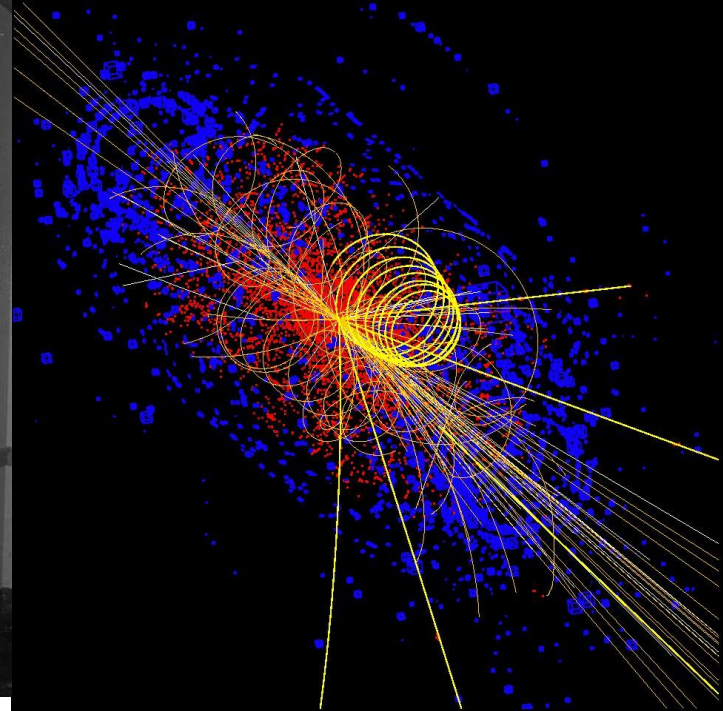
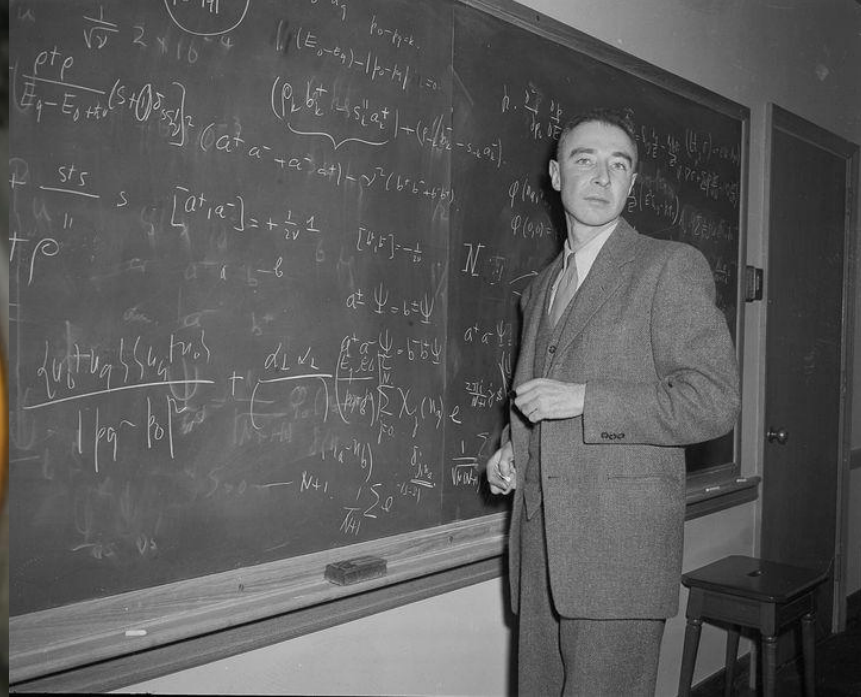
1. Bound state solution : Attractive potential
2. Scattering solution : Repulsive potential.

# Bound states

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- Physically, a bound state can be viewed as a “trapped” scattering state whose outgoing component has disappeared, leaving only a decaying solution; mathematically, this is why it emerges as a pole of the scattering amplitude under  $k \rightarrow i\kappa$
- Low-energy scattering is strongly influenced by near-threshold bound states; in particular, when a bound state is close to zero energy, the phase shift varies rapidly and the scattering length becomes large, reflecting long-range sensitivity of the wavefunction
- The entire structure shows that bound states and scattering are not separate phenomena but two analytic continuations of the same radial equation, with the phase shift acting as the bridge between them





THANK YOU FOR  
YOUR TIME!!

Questions?

